

in such vast numbers that the sea appears milky with gametes. The sperm quickly fertilize the floating eggs, and within hours the palolo's once-a-year reproductive frenzy is complete.

When external fertilization is not synchronous across a population, individuals may exhibit specific “courtship” behaviors leading to the fertilization of the eggs of one female by one male (see Figure 46.5). By triggering the release of both sperm and eggs, these behaviors increase the probability of successful fertilization.

Internal fertilization is an adaptation that enables sperm to reach an egg even when the external environment is dry. It typically requires sophisticated and compatible reproductive systems, as well as cooperative behavior that leads to copulation. The male copulatory organ delivers sperm, and the female reproductive tract often has receptacles for storage and delivery of sperm to mature eggs.

No matter how fertilization occurs, the mating animals may make use of *pheromones*, chemicals released by one organism that influence the physiology and behavior of other individuals of the same species. Pheromones are small, volatile or water-soluble molecules that disperse into the environment and are active at very low concentrations (see Concept 45.1). Many pheromones function as mate attractants. For example, pheromones allow some female insects to be detected by males more than a kilometer away.

Evidence for human pheromones remains controversial. It was once argued that female roommates produce pheromones that trigger synchrony in menstrual cycles, but further statistical analyses have failed to support this finding.

## Ensuring the Survival of Offspring

Typically, animals that fertilize eggs internally produce fewer gametes than species with external fertilization, but a higher fraction of their zygotes survive. Better zygote survival is due in part to the fact that eggs fertilized internally are sheltered from potential predators. However, internal fertilization is also more often associated with mechanisms that provide greater protection of the embryos and parental care of the young. For example, the internally fertilized eggs of birds and other reptiles have shells and internal membranes that protect against water loss and physical damage during the eggs' external development (see Figure 34.26). In contrast, the eggs of fishes and amphibians have only a gelatinous coat and lack internal membranes.

Rather than secreting a protective eggshell, some animals retain the embryo for a portion of its development within the female's reproductive tract. The offspring of marsupial mammals, such as kangaroos and opossums, spend only a short period in the uterus as embryos; they then crawl out and complete development attached to a mammary gland in the mother's pouch. Embryos of eutherian (placental) mammals, such as zebras and humans, remain in the uterus throughout fetal development. There they are nourished by the mother's blood supply through a temporary organ, the placenta. The embryos of some fishes and sharks also complete development internally.

▼ **Figure 46.6 Parental care in an invertebrate.** Compared with many other insects, giant water bugs of the genus *Belostoma* produce relatively few offspring but offer much greater parental protection. Following internal fertilization, the female glues her fertilized eggs to the back of the male. The male (shown here) carries the eggs for days, frequently fanning water over them to keep them moist, aerated, and free of parasites.



When a caribou or kangaroo is born or when a baby eagle hatches out of an egg, the newborn is not yet capable of independent existence. Instead, mammals nurse their offspring, and adult birds feed their young. Parental care of eggs or offspring is in fact widespread among animals, even including some invertebrates (Figure 46.6).

## Gamete Production and Delivery

Sexual reproduction in animals relies on sets of cells that are precursors for eggs and sperm. Cells dedicated to this function are often established early in the formation of the embryo and remain inactive while the body plan takes shape. Cycles of growth and mitosis then increase, or *amplify*, the number of cells available for making gametes—eggs or sperm.

In producing gametes from the precursor cells and making them available for fertilization, animals employ a variety of reproductive systems. **Gonads**, organs that produce gametes, are found in many but not all animals. Exceptions include the palolo, discussed above. The palolo and most other polychaete worms (phylum Annelida) have separate sexes but lack distinct gonads; rather, the eggs and sperm develop from undifferentiated cells lining the coelom (body cavity). As the gametes mature, they are released from the body wall and fill the coelom. Depending on the species, mature gametes in these worms may be shed through the excretory opening, or the swelling mass of eggs may split open a portion of the body, spilling the eggs into the environment.

More elaborate reproductive systems include sets of accessory tubes and glands that carry, nourish, and protect the gametes and sometimes the developing embryos. For example, fruit flies and most other insects have separate sexes with